

Wash and Rack Dishes SOP

One episode washes three dishes. It runs from three dirty dishes standing at the dirty spot, a wash basin of soapy water, a rinse basin of clean water, a bare dish rack and a dry cloth on its spot, to three clean dishes standing in the rack, each one in the slot marked for its own type, the counter around the basins wiped dry, and every tool back on its spot.

The order never changes: **scrub, rinse, drain, rack**. One dish goes all the way through before the next dish is picked up. The dishes are worked in one fixed order: the plate first, then the bowl, then the cup. The splashes are wiped last, after all three dishes stand in the rack.

One gripper holds and the other gripper works. The holding gripper is closed and empty and presses the dish down so it stays still. The working gripper picks up, carries, scrubs and places. The two never trade jobs part way through a step. Nothing is ever passed from one gripper to the other, only one dish is carried at a time, and no grasp is changed while a dish is off the surface.

The two basins stand side by side across the midline and are shared, so each gripper meets a basin from its own side. Nothing is carried across to the far side of the counter, and no carry passes over the dish rack, over the dirty dishes or over the brush.

The counts are fixed: **three dishes, six brush strokes on each dish, one dip in the rinse basin, one drain of 3 seconds, and three racked dishes**. There is no second scrub and no second dip. A dish is never washed twice. Water that lands on the counter is wiped in Step 5. Water or a dish that lands on the floor is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole counter in one shot: the dirty spot, the brush spot, the wash basin, the rinse basin, the splash strip, the dish rack and the cloth spot.
3. The inside of each basin is in frame, so the dish is seen while it is scrubbed and while it is

dipped. The smear on a dish is visible before the scrub, and all three rack slots stay in frame for the whole episode.

4. Both arms are at home with their grippers open.
5. The counter is dry and clear of anything but the items in the Materials checklist.
6. Each arm reaches the dirty spot, the brush spot, both basins, the whole splash strip, all three rack slots and the cloth spot without extending to a joint limit or folding in on itself. Bring a spot forward until this is true.
7. Each arm reaches both basins from its own side of the midline. Neither has to lean over the other to work them.
8. Neither arm needs the far side of the counter for anything. The paths from the dirty spot to the wash basin, from the wash basin to the rinse basin, and from the rinse basin to the rack are clear. Carry one dish along each path by hand during setup: it must clear both basin rims and pass over no other dish. Widen the gaps until it does.

Materials checklist

1. Three dirty dishes stand side by side at the dirty spot, in the order they are worked, left to right: one plate, one bowl, one cup. They do not touch each other.
2. The three dishes are three different types, and each type matches one slot in the rack. The same three types are used in every episode.
3. Each dish is dry on the outside at the start and carries one food smear on its inside face. The smear is one patch, plain to see on camera, and it comes off under six strokes of the brush.
4. No dish carries baked on food, and no dish is chipped or cracked. Each dish has a rim the gripper can close on squarely, and hangs level from that grip.
5. Each dish is light enough for one gripper to carry level when it is wet, and small enough to lie flat on the floor of a basin with nothing standing above the basin rim.
6. One wash basin stands on its spot, holding warm soapy water to about half its depth. The water covers a dish lying in it, and it is clear enough that the dish and the smear are still seen through it.
7. One rinse basin stands beside it, holding clean water to about half its depth, deep enough to cover a dish and clear at the start of every episode.
8. Both basins are open at the top and wide enough that a dish goes in without being pushed. Push each basin by hand during setup: if it slides or lifts, weigh it down. Nothing pins a basin during the episode.
9. One brush lies flat on the brush spot, bristles down and handle toward the front. It is light, and its handle is long enough that the gripper stays above the water while the bristles reach the dish.
10. One dish rack stands empty on the rack spot with three slots, one for each type: the plate slot, the bowl slot and the cup slot. Each slot is marked so the reviewer can tell the three

apart on camera.

11. Each slot is wide enough that its dish goes in without being pushed and holds that dish still once it is in. The rack sits flat and does not rock.
12. One dry cloth, folded flat, lies on the cloth spot. It is clean and dry, and it takes up water without dripping.
13. The splash strip, the bare counter between the basins and the rack, is dry and clear at the start.

Workspace layout

Three rows: the dirty dishes and the brush at the back, the two basins in the middle, the dish rack at the front with the cloth beside it. Everything travels forward, from the dirty spot to the rack.

Back row, left to right:

1. Dirty spot, back left. The three dirty dishes stand side by side here, the plate, then the bowl, then the cup. It empties as the rack fills.
2. Brush spot, back right. The brush lies flat here, bristles down and handle toward the front.

Middle row, left to right:

3. Wash basin, just left of the midline, holding the soapy water. Shared: one gripper holds the dish in it and the other works the brush, and each meets the basin from its own side.
4. Rinse basin, just right of the midline, holding the clean water. Shared in the same way. The dish is dipped here and drained here.

Front row, left to right:

5. Splash strip, the band of bare counter between the basins and the rack. Dry at the start, wiped dry again at the end.
6. Dish rack, front center, its three marked slots facing up. Empty at the start and holding three clean dishes at the end. It is the output of the episode.
7. Cloth spot, front right beside the rack.

A clear gap of at least a gripper's width sits between the two basins, and between the rinse basin and the rack, so a dish lifts straight up and travels forward without touching anything. No carry passes over the dirty dishes, over the brush, or over any rack slot other than the one the dish is going into.

Vocabulary

- **Hold:** a closed empty gripper pressed onto a dish to keep it still. A hold never lifts, carries or moves what it holds.

- **Work:** the other gripper picking up, carrying, scrubbing or placing. One gripper holds and the other works, and the two never trade jobs part way through a step.
- **Tool home:** the spot a tool returns to and lies flat on before the next step starts: the brush spot and the cloth spot. A tool is never left in a basin, in the rack, on the splash strip or in a gripper between steps.
- **Release point:** the gripper opens over a basin when a dish is being put in, or over the slot that dish belongs in, or over a tool home. Never over another slot, over the dirty spot or over open counter.
- **Stable placement:** the dish stands still after release, on its own surface, and does not rock, tip, slide or lean on anything.
- **Dirty dish:** a dish carrying one food smear on its inside face. Every episode starts with three, one of each type.
- **Smear:** the patch of soft food on the inside face of a dish. It is one patch, plain to see on camera, and it comes off under the brush.
- **Clean dish:** no smear left anywhere on the dish and no food still stuck to it.
- **Brush:** the handled brush that takes the smear off. Only its bristles go under the water. The gripper stays above the water.
- **Stroke:** one draw of the brush across the dish, bristles flat and in contact the whole way, from one side of the dish to the other.
- **Scrubbed:** six strokes made across the smear, overlapping, with no smear left on the dish.
- **Dip:** the dish lowered straight down into the rinse water until it is fully under, held still, then lifted straight out. One dip a dish.
- **Rinsed:** no soap foam left anywhere on the dish after the dip.
- **Drained:** the dish held over a basin until the water stops running off it and only single drops fall.
- **Dish rack:** the rack the clean dishes stand in. It is the output of the episode.
- **Slot:** one marked place in the rack that takes one type of dish: the plate slot, the bowl slot, the cup slot.
- **By type:** each dish goes in the slot marked for its own type. A dish in the wrong slot is a wrong placement, even if it stands well.
- **Splash:** water on the counter around the basins or on the splash strip. It is part of the task and is wiped in Step 5.
- **Spill:** water or a dish on the floor, or water off the counter edge. It is left where it falls and cleared in the reset, never on camera.
- **Cloth:** the folded dry cloth used on the splash strip. It is held so a broad flat side meets the counter, never by a corner and never balled up. It touches the counter and nothing else, ever.
- **Blot:** the cloth pressed flat on a wet patch and held still so the water soaks in, then lifted

straight up. The cloth is never dragged along the counter: dragging spreads the water instead of taking it up.

- **Quarter turn:** the most any dish or tool is tilted in a gripper. Standing a plate in its slot and draining a dish both stay inside it. Nothing is ever turned over in the air.

Steps

Steps 1 to 4 run three times, once for each dish, and one dish goes all the way through before the next one is picked up. The plate is worked first, then the bowl, then the cup.

Step 1: Scrub one dish in the wash basin

Goal: the dish lies in the wash basin with no smear left on it, and the brush is back on its spot.

The dish is scrubbed lying in the basin, so it is never held in the air while the brush works on it.

1.1 Put the dish in the wash basin

- With one gripper, close on the rim of the next dirty dish, square to it so it hangs level. The rim is the grasp: the grippers cannot span a dish wall securely.
- Lift it straight up clear of the counter, carry it forward level without tilting or swinging it, and keep it clear of the other dirty dishes.
- Lower it into the soapy water until it rests on the basin floor, smear face up, then open the gripper and lift clear.

Check: the dish lies flat on the basin floor with the smear facing up, and no part of it stands above the basin rim. If it leans on a basin wall, lift it clear and lay it down again. Make no more than one correction.

1.2 Hold the dish and take the brush

- With one gripper closed and empty, come down on the dish and press it against the basin floor. Take this hold **before the brush goes into the water**, and keep it **until the sixth stroke is done**.
- With the other gripper, close on the brush handle end, lift it straight up off the brush spot, and carry it to the wash basin.

A dish in water is light and slides under the brush, so an unheld dish moves away from the bristles and the smear is never reached.

1.3 Scrub, six strokes

- Lower the brush until the bristles meet the smear, bristles flat on the dish.
- Draw the brush straight across the smear from one side of the dish to the other, in contact

the whole way.

- Lift the bristles clear of the dish at the end of each stroke and bring them back through the water to the start side. Never draw the brush back across the dish.
- Step each stroke beside the last, overlapping it, and work across until the whole smear has been covered.
- Make **six strokes in all**. Nothing else changes between them.

Only the bristles go under the water. The gripper stays above it for the whole scrub.

Check: no smear is left on the dish. A dish that still shows a smear after six strokes is left as it is: do not scrub again, and let review flag the episode.

1.4 Return the brush to its spot

- Carry the brush back to the brush spot, lay it flat bristles down, and release.
- The holding gripper releases the dish and lifts clear only now.

Check: the brush lies fully on its spot, not in a basin, not on the rack and not on the splash strip.

Step 2: Rinse the dish

Goal: the dish dipped once in the rinse basin, with no soap foam left on it and the dish still in the gripper.

2.1 Move the dish to the rinse basin

- With one gripper, close on the dish rim square, lift it straight up out of the soapy water, and hold it still over the wash basin for **2 seconds** so the first water runs back in.
- Carry it level to the rinse basin and stop above the rinse water, inside the ring of the basin rim.

The dish never passes over the dish rack, over the dirty dishes or over the brush.

2.2 Dip

- Lower the dish straight down into the rinse water until it is fully under.
- Hold it still under the water for **2 seconds**.
- Lift it straight up out of the water, keeping it level. Do not let go of it in the rinse basin.

One dip, and one only. A dish that still carries foam is tagged and the episode continues. There is no second dip and no wipe with the cloth.

Check: no soap foam is on the dish, the dish is still held in the gripper, and the rinse water is still in its basin.

Step 3: Drain the dish

Goal: the water stops running off the dish, and every drop runs back into the rinse basin.

- Hold the dish still above the rinse basin, inside the ring of the basin rim and low to it.
- Tilt it **no more than a quarter turn** so the water runs off one edge, and hold it there for **3 seconds**.
- Bring the dish back to level before it leaves the basin mouth.

The dish is drained over the basin and nowhere else. A dish carried to the rack while it is still running lays water on the splash strip and into the rack.

Check: only single drops fall from the dish. If water still runs off it in a stream, hold it over the basin for 3 seconds more. Make no more than one extra drain.

Step 4: Rack the dish by type

Goal: the dish standing still in the slot marked for its own type, clear of the other slots.

- Carry the dish forward and level to the rack and stop above the slot marked for its type: the plate over the plate slot, the bowl over the bowl slot, the cup over the cup slot.
- Lower it into that slot. Stand the plate on its edge, turning it **no more than a quarter turn**. Lower the bowl mouth up. Lower the cup mouth up. Nothing is turned over in the air.
- Release once the dish stands on its own, then lift clear.
- Leave the other slots clear. The dish does not lean on another dish and does not overhang the rack.

Check: the dish stands still in its own slot, upright and clear of the others. If it is in the wrong slot, rocks, or leans on another dish, lift it clear and place it again. Make no more than one correction per dish.

Now go back to Step 1 with the next dish. Steps 1 to 4 run three times in all: the plate, then the bowl, then the cup. Start Step 5 only when all three dishes stand in the rack.

Step 5: Wipe the splashes

Goal: the splash strip is dry and the cloth is back on its spot.

5.1 Take the cloth

- Come down on the folded cloth on its spot from above and close on it so a broad flat side hangs below the gripper.

Check: the cloth shows a flat face and does not hang by a corner or slip. If it does, lay it back on its spot and take it again. Make no more than two corrective takes. If it still will not sit flat, end the episode and correct the cloth during reset.

5.2 Blot the wet patches

- Bring the cloth down flat onto one wet patch on the splash strip, press it against the counter, and hold it still for **2 seconds** so the water soaks in.
- Lift the cloth straight up off the patch. Never drag it along the counter.
- Move to the next wet patch and blot it the same way. Work left to right along the strip until every wet patch has been blotted.
- Blot each patch once. A patch that is still wet takes one more blot and no more.

The cloth touches the counter and nothing else. It never goes into a basin, onto the rack, or onto a dish.

Check: the splash strip is dry and no wet patch is left on it. Water on the floor is left where it is and cleared in the reset.

5.3 Return the cloth to its spot

- Carry the cloth back to the cloth spot, lay it flat, release, and lift clear.

Check: the cloth lies fully on its spot, not draped over the rack, a basin or the counter edge.

Step 6: End the episode

- Confirm the dishes: all three in the rack, each in the slot for its own type, each standing still, and none of them touching another.
- Confirm the dirty spot: bare.
- Confirm the basins: both still on their spots, both still holding their water, and no dish left in either of them.
- Confirm the splash strip: dry.
- Confirm the tools: the brush flat on its spot and the cloth flat on its spot.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return both arms home with their grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the three dishes out of the rack, wash and dry them by hand, and put one fresh food smear on the inside face of each one, one patch to a dish, plain to see on camera.
2. Stand the three dishes at the dirty spot side by side in working order, the plate, then the bowl, then the cup, not touching each other.
3. Tip out both basins, wash them out, and fill them again: warm soapy water in the wash

basin and clean water in the rinse basin, each to about half its depth.

4. Rinse the brush, shake it dry, and lay it flat on the brush spot, bristles down and handle toward the front. Replace a brush whose bristles are worn flat.
5. Dry the whole counter, the splash strip and the rack, and stand the rack empty on its spot with all three slots clear and their marks easy to read.
6. Pick up anything that fell on the floor and dry the floor around the cell.
7. Lay a fresh folded dry cloth flat on the cloth spot. Replace a cloth that is damp or dirty.
8. Confirm no dish is chipped or cracked, both basins still stand still and do not leak, and the rack sits flat and does not rock.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a dish goes into the rinse basin before it has been scrubbed, a dish goes into the rack before it has been dipped or before it has been drained, the cloth comes out before all three dishes stand in the rack, or a step is skipped.
- **SOP rule broken:** Steps 1 to 5, scrub, rinse, drain, rack, then wipe, in that order.
- **Coaching note:** scrub it, dip it, drain it, rack it. The cloth is the last thing that moves.

Violation: Dishes worked out of order, or two dishes in play at once

- **Visible cue:** the bowl or the cup is picked up before the plate is in the rack, a dish is lifted off the dirty spot while another dish is still in a basin, or two dishes are carried in one trip.
- **SOP rule broken:** Steps 1 to 4, one dish goes all the way through before the next is picked up, in the fixed order plate, bowl, cup.

- **Coaching note:** one dish at a time, all the way to the rack. Plate, bowl, cup, every episode.

Violation: Dish not held, or scrubbed loose

- **Visible cue:** the holding gripper is off the dish at any point between the brush entering the water and the sixth stroke; the dish slides, turns or floats up under the brush; or the dish is lifted out of the basin while the brush is still on it.
- **SOP rule broken:** Steps 1.2 and 1.3, the dish is pressed against the basin floor until the sixth stroke is done, and it is scrubbed lying in the basin.
- **Coaching note:** the holding gripper holds through all six strokes. The dish is never scrubbed in the air.

Violation: Scrub count wrong

- **Visible cue:** fewer or more than six strokes are made on a dish, or a seventh stroke is made to take off a smear the first six missed.
- **SOP rule broken:** Step 1.3, six strokes in all on each dish.
- **Coaching note:** the count is fixed at six. A correction stroke makes the episode unlearnable.

Violation: Stroke wrong

- **Visible cue:** the bristles lift part way through a stroke, a stroke stops short of the other side of the dish, strokes do not overlap, parts of the smear are never touched, the brush is drawn back across the dish still in contact with it, or the brush is dabbed or scrubbed back and forth instead of drawn in single strokes.
- **SOP rule broken:** Step 1.3, overlapping strokes in contact from one side of the dish to the other, with the bristles lifted clear and brought back through the water.
- **Coaching note:** one draw across, lift, come back through the water. The back stroke is the most common reason a smear stays on.

Violation: Gripper under the water

- **Visible cue:** the gripper holding the brush goes below the water line, or the brush is let go of in a basin.
- **SOP rule broken:** Step 1.3, only the bristles go under the water and the gripper stays above it.
- **Coaching note:** the handle is long for a reason. Keep the gripper in the air.

Violation: Rinse done wrong

- **Visible cue:** the dish is not fully under the rinse water, it is held under for less than 2 seconds, it is let go of in the rinse basin, it is dipped a second time after it has come out, or it goes to the rack with soap foam still on it and no dip made.

- **SOP rule broken:** Step 2.2, one dip, fully under the water, held 2 seconds, lifted straight out and never released in the basin.
- **Coaching note:** all the way under, hold, straight out. One dip, and one only.

Violation: Dish racked wet, or drained in the wrong place

- **Visible cue:** water still runs off the dish in a stream as it leaves the rinse basin, the dish is tilted to drain while it is over the splash strip or over the rack, or the drain is held for less than 3 seconds.
- **SOP rule broken:** Step 3, the dish is drained over the rinse basin for 3 seconds until only single drops fall.
- **Coaching note:** drain over the basin, not on the way. Wait for the stream to stop before the carry starts.

Violation: Dish in the wrong slot

- **Visible cue:** a dish ends in a slot that is not marked for its type, two dishes end in one slot, a dish leans on the dish beside it, or part of a dish hangs off the rack.
- **SOP rule broken:** Step 4, each dish stands in the slot marked for its own type, clear of the other dishes and fully on the rack.
- **Coaching note:** read the slot mark before the dish goes down. Plate to plate, bowl to bowl, cup to cup.

Violation: Dish released before it is stable

- **Visible cue:** a gripper opens while the dish is still moving, or the dish rocks, tips or slides after release, in the rack or in a basin.
- **SOP rule broken:** Steps 1.1 and 4, the dish stands still after release and does not rock.
- **Coaching note:** let the dish take its own weight before the gripper opens.

Violation: Cloth held wrong, or the counter wiped without it

- **Visible cue:** the cloth is held by a corner or balled up so only a narrow line of it meets the counter, or the counter is wiped with the bare gripper.
- **SOP rule broken:** Step 5.1, the cloth is held so a broad flat side meets the counter.
- **Coaching note:** a broad flat face on the counter. A bare gripper spreads the water instead of taking it up.

Violation: Cloth dragged, or splashes left behind

- **Visible cue:** the cloth is drawn along the counter instead of pressed and lifted, water is pushed toward the front edge or off it, a patch is blotted more than twice, a wet patch is left on the splash strip at the end, or the cloth goes into a basin, onto the rack or onto a dish.

- **SOP rule broken:** Step 5.2, each wet patch is blotted once, pressed for 2 seconds and lifted straight up, and the cloth touches the counter and nothing else.
- **Coaching note:** press, wait, lift. A dragged cloth makes one small patch into a big one.

Violation: Spill cleaned up

- **Visible cue:** after water or a dish lands on the floor or off the counter edge, a gripper or the cloth is used to wipe, push or pick it up.
- **SOP rule broken:** Steps 1 to 5, a spill off the counter is left where it falls and is cleared in the reset. Only the splash strip is wiped, and only in Step 5.
- **Coaching note:** the splash strip is the task. The floor is not. The spill is data, the cleanup is noise.

Violation: Hold and work traded part way

- **Visible cue:** the gripper holding the dish takes the brush, the gripper working the brush takes over the hold, or the two swap jobs in the middle of a scrub.
- **SOP rule broken:** Steps 1.2 and 1.3, one gripper holds and the other works, and the two never trade jobs part way through a step.
- **Coaching note:** pick the roles at the start of the step and keep them to the end of it.

Violation: A carry passed over the rack, the dirty dishes or the brush

- **Visible cue:** a wet dish passes over a rack slot other than the one it is going into, a dish passes over the dirty dishes or over the brush, or the brush passes over the rack or over the dirty dishes.
- **SOP rule broken:** Steps 1 to 4, no carry passes over the dirty dishes, over the brush, or over any rack slot other than the one the dish is going into.
- **Coaching note:** a wet dish drips on whatever it crosses. Route around, not across.

Violation: Bad pick, carry, or release

- **Visible cue:** a dish moves from one gripper to the other without being set down first, both grippers hold one dish at once, a gripper shifts or seats its hold again while the dish is off the surface, a dish or the brush slides along the counter with no daylight under it, a dish hangs tilted in the gripper because the rim grasp was taken off square, or a dish is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 5, one object at a time, lifted straight up clear of the surface before any sideways motion, gripped square on the rim, tilted no more than a quarter turn, and never passed between grippers.
- **Coaching note:** set it down to change the grip, and lift before the carry. A dish that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a dish, a basin, the rack or a tool is shoved, nudged or slid into place instead of being lifted clear and placed again, a dish is pushed down into a slot that is too tight, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 5, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** the brush is left in a basin, on the rack, on the splash strip or in a gripper when the next step begins; the cloth is left on the counter, on the rack or over a basin; or a gripper opens while the tool is still above its spot.
- **SOP rule broken:** Steps 1.4 and 5.3, each tool returns to its home and lies flat there before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower until it touches, then release.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, a gripper or a carried dish strikes a basin, the rack, the brush or another dish and knocks, tips, slides or displaces it; the two arms strike each other over a basin; a basin is moved off its spot; or a dish falls short of its slot, with no gripper failure, drift or controller fault visible.
- **SOP rule broken:** Steps 1 to 5, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. On the basin, the working gripper moves and the holding gripper stays put.

Violation: Manipulation outside the camera frame

- **Visible cue:** a scrub, a dip, a drain, a placement or a blot happens partly or wholly outside the environment camera frame.
- **SOP rule broken:** Steps 1 to 5, every manipulation happens inside the camera frame.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a dish still at the dirty spot or still in a basin, a dish in the wrong slot, the splash strip still wet, a tool out of its home, an arm short of home, or a gripper not fully open.

- **SOP rule broken:** Step 6, confirm the dishes, the dirty spot, the basins, the splash strip and the tools, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside an arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective basin: one that slides or lifts under a correct scrub, that leaks, that is too deep to see the dish in, or that is too narrow for a dish to go in without being pushed. Weigh it down or replace it before the next episode.
- Defective dish: one that is chipped or cracked, that will not lie flat on a basin floor, or whose rim the gripper cannot close on squarely.
- Defective brush: one whose bristles are worn flat, whose handle is too short to keep the gripper above the water, or that sheds bristles into a basin.
- Defective rack: one that rocks, one whose slot marks cannot be read on camera, or one with a slot too narrow for its dish.
- Damp or dirty cloth at episode start, or a cloth that drips instead of taking up water.
- Baked on food or a hard smear at episode start, so six strokes cannot take it off any dish. Reset error, not a violation.
- A basin holding too little water to cover a dish, or rinse water already cloudy with soap at episode start. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Lift one dirty dish from the dirty spot
2. Put one dish in the wash basin
3. Hold one dish steady in the basin
4. Take the brush from its spot
5. Scrub one stroke across the dish
6. Return the brush to its spot
7. Lift one dish out of the wash basin

8. Dip one dish in the rinse basin
9. Drain one dish over the rinse basin
10. Place one dish in its slot in the rack
11. Take the cloth from its spot
12. Blot one wet patch on the counter
13. Return the cloth to its spot
14. Return both arms home and end the episode